



SQL Bookstore Project

An Online Bookstore Database Management System built with PostgreSQL



Project Overview

An **Online Bookstore Database Management System** developed using **PostgreSQL** — designed to simulate real-world data management for a functioning online bookstore.

Relational Schema

3 interconnected tables
designed with proper
constraints and relationships

Data Import

CSV files loaded into
PostgreSQL using the COPY
command

20 SQL Queries

Real-world business problems solved across sales, customers, books,
and inventory



Database Structure



Books

- Book_ID **(PK)**
- Title, Author, Genre
- Published_Year, Price, Stock



Customers

- Customer_ID **(PK)**
- Name, Email, Phone
- City, Country



Orders

- Order_ID **(PK)**
- Customer_ID **(FK)**, Book_ID **(FK)**
- Order_Date, Quantity, Total_Amount

Table Relationships

The three tables are linked through **Foreign Key** relationships, forming the backbone of the relational schema.

- One customer can place **multiple orders**
- One book can appear in **multiple orders**
- **Customer_ID** and **Book_ID** serve as Foreign Keys in the Orders table

Table	Columns
Books	Book_ID, Title, Author, Genre, Published_Year, Price, Stock
Customers	Customer_ID, Name, Email, Phone, City, Country
Orders	Order_ID, Customer_ID (FK), Book_ID (FK), Order_Date, Quantity, Total_Amount



Project Workflow

1

Create Database

Set up the OnlineBookstore PostgreSQL database

2

Create Tables

Designed Books, Customers, and Orders tables with proper constraints

3

Import Data

Loaded Books.csv, Customers.csv, and Orders.csv using the COPY command

4

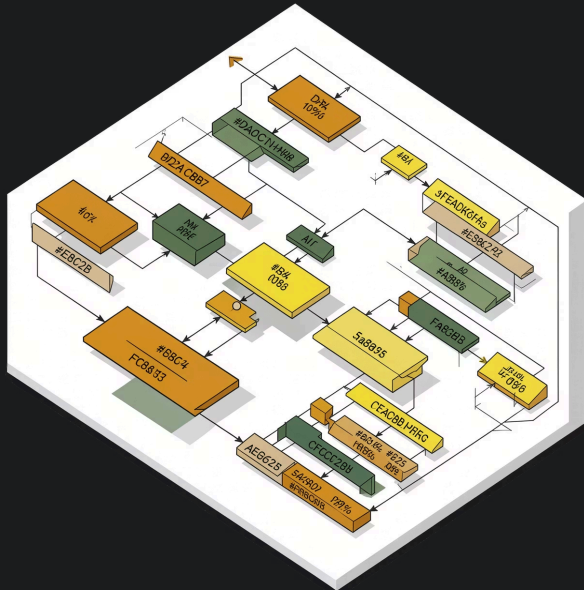
Write Queries

Solved 20 real-world SQL business problems





SQL Concepts Used



Filtering & Sorting

SELECT, WHERE, ORDER BY,
DISTINCT, LIMIT

Grouping & Aggregation

GROUP BY, HAVING, COUNT,
SUM, AVG

Joins

INNER JOIN, LEFT JOIN across
relational tables

Other Techniques

COALESCE for handling NULL
values gracefully



Business Problems Solved (1–10)

1

Fiction Books

Retrieve all books in the Fiction genre

2

Post-1950 Books

Find books published after 1950

3

Canadian Customers

List all customers from Canada

4

Nov 2023 Orders

Show orders placed in November 2023

5

Total Stock

Calculate total available inventory stock

1

Priciest Book

Find the most expensive book in the catalog

2

Multi-Book Buyers

Find customers who ordered more than one book

3

High-Value Orders

Retrieve orders with total amount > \$20

4

All Genres

List all distinct available book genres

5

Low Stock Alert

Find the book with the lowest remaining stock



Business Problems Solved (11–20)



→ Total Revenue

Calculate overall revenue generated from all orders

→ Books Sold by Genre

Aggregate quantity sold broken down by genre

→ Repeat Customers

Find customers with at least two orders placed

→ Top Ordered Book

Identify the most frequently ordered book

→ Top Fantasy Books

Show the top 3 most expensive Fantasy books

→ Author Sales & Top Spender

Books sold per author, highest spending customer, cities spending >\$30, and remaining stock after orders

Skills Demonstrated



Database Design

Relational schema with Primary Keys & Foreign Keys



Data Import

CSV ingestion via PostgreSQL COPY command



SQL Querying

Joins, aggregations, filtering, and subqueries



Business Analysis

Sales, inventory, and customer insights





Project Outcome

3

Relational Tables

Books, Customers, and Orders — fully linked with PKs & FKs

3

CSV Files Imported

Data loaded seamlessly using the PostgreSQL COPY command

20

Business Problems

Real-world SQL queries solved across sales, inventory, and customers

- ✓ Successfully designed a fully relational Online Bookstore database — from schema creation to solving 20 real-world SQL business problems using PostgreSQL.



About the Author



Ajit Rana

Aspiring Data Analyst | Open to Work

Technical Skills



Advanced Excel

Data analysis, pivot tables, and dashboards



SQL

PostgreSQL querying, schema design, and data management



Power BI

Interactive reports and business intelligence dashboards



Python

Data wrangling, analysis, and automation scripts

